(CS-4-AI-1) Roll No. .....

PAPER ID—21134

B.Tech. EXAMINATION, 2024

(Artificial Intelligence and Machine Learning)

(Fourth Semester)

COMPUTER SCIENCE AND ENGINEERING

Operating System

Time : 3 *Hours* *Maximum Marks* : 70

Before answering the question-paper candidates should ensure that they have been supplied to correct and complete question-paper. No complaint, in this regard, will be entertained after the examination.

Note : Attempt *Five* questions in all, selecting *one* question from each Unit. Q. No. **1** is compulsory. All questions carry equal marks.

1. (a) What is a Semaphore ?

(b) What is a Thread ?

(c) Make a diagram of PCB.

(d) Describe Race Condition.

(e) What is a Scheduler ? What is its purpose ?

(f) Define SCAN algorithm.

(g) What is Shell in Unix ? 7×2=14

Unit I

2. (a) Define a process in the context of operating systems. What are the different states a process can be in ? 7

(b) Consider a system with four processes P1, P2, P3 and P4. Their CPU burst times are 10 ms, 5 ms, 8 ms, and 12 ms, respectively. Calculate the average turnaround time using : 7

(i) First-Come, First-Served (FCFS) scheduling.

(ii) Shortest Job First (SJF) scheduling.

3. (a) Discuss the Round-Robin scheduling algorithm. How does it work, and what is the significance of the time quantum parameter ? 7

(b) Consider a system using feedback scheduling with three queues : Q1, Q2, and Q3. The time quantums for Q1, Q2, and Q3 are 4 ms, 8 ms, and 16 ms, respectively. If a process in Q1 moves to Q2 after exhausting its time quantum and then to Q3 after exhausting its time quantum in Q2, and each queue has one process with CPU burst times of 12 ms, 10 ms and 15 ms, respectively, calculate the average turnaround time. 7

Unit II

4. (a) Describe two methods used to achieve mutual exclusion in concurrent programs, and compare their advantages and disadvantages. 7

(b) Explain the difference between deadlock prevention and deadlock avoidance strategies. 7

5. (a) Discuss the role of mutual exclusion in preventing race conditions and ensuring program correctness. 7

(b) Describe the Banker's algorithm for deadlock avoidance. How does it ensure safe state allocation of resources ? 7

Unit III

6. (a) Explain the concept of locality of reference in the context of memory management. How does locality of reference influence the effectiveness of caching and page replacement algorithms ? 7

(b) A computer system employs demand paging with a page size of 4 kB. If a process requires 32 kB of memory and the system has 128 kB of physical memory, calculate the following : 7

(i) The number of pages required to store the process.

(ii) If the process accesses pages in the following order :

1, 2, 3, 4, 1, 5, 6, 2, 7,

calculate the page faults using FIFO and LRU page replacement algorithms.

7. Compare and contrast contiguous memory allocation with paged memory allocation. Discuss the advantages and disadvantages of each approach in memory management. 14

Unit IV

8. (a) Compare and contrast FCFS and SSTF disk scheduling algorithms. Provide scenarios where each algorithm would be suitable. 7

   (b) Compare and contrast FAT and NTFS. 7

9. Discuss the various disk scheduling algorithms. Compare their performance and suitability for different disk access patterns. 14